

CS228: Assignment 1

Report and Analysis

Wing Lee

Introduction:

I used Gait recognition which is a biometric technique that looks for individuals based on their patterns. In this assignment we used a user id and verification system using data collected from accelerometer and gyroscope signals. The dataset uses a 6 channel signal that uses different user movements. We use two different Machine learning approaches first a, Forward Neural Networks (FNN) was used for ID based on signal features and extracted from window sensor data. Feature selection was used to decide the most important subset of features before training the classifier. Second, we used the Long Short Term Memory (LSTM) to learn patterns directly from the data. Unlike the FNN approach, LSTM processes the signal over time instead of using extracted features.

Both models were evaluated on two tasks: the user id and the user verification, binary classification was used to distinguish genuine users from imposters. We then use FAR, FRR and EER and also the ROC to evaluate the metrics.

Preprocessing:

The data have 6 sensor signals, a three axis accelerometer and a three axis gyroscope both recorded across three axes (X, Y, Z) for 10 distinct users during gait activities. I combined the 3 accelerometer and 3 gyroscope into a single 6 channel. Then I windowed the data for 500 window size and 50% overlap.

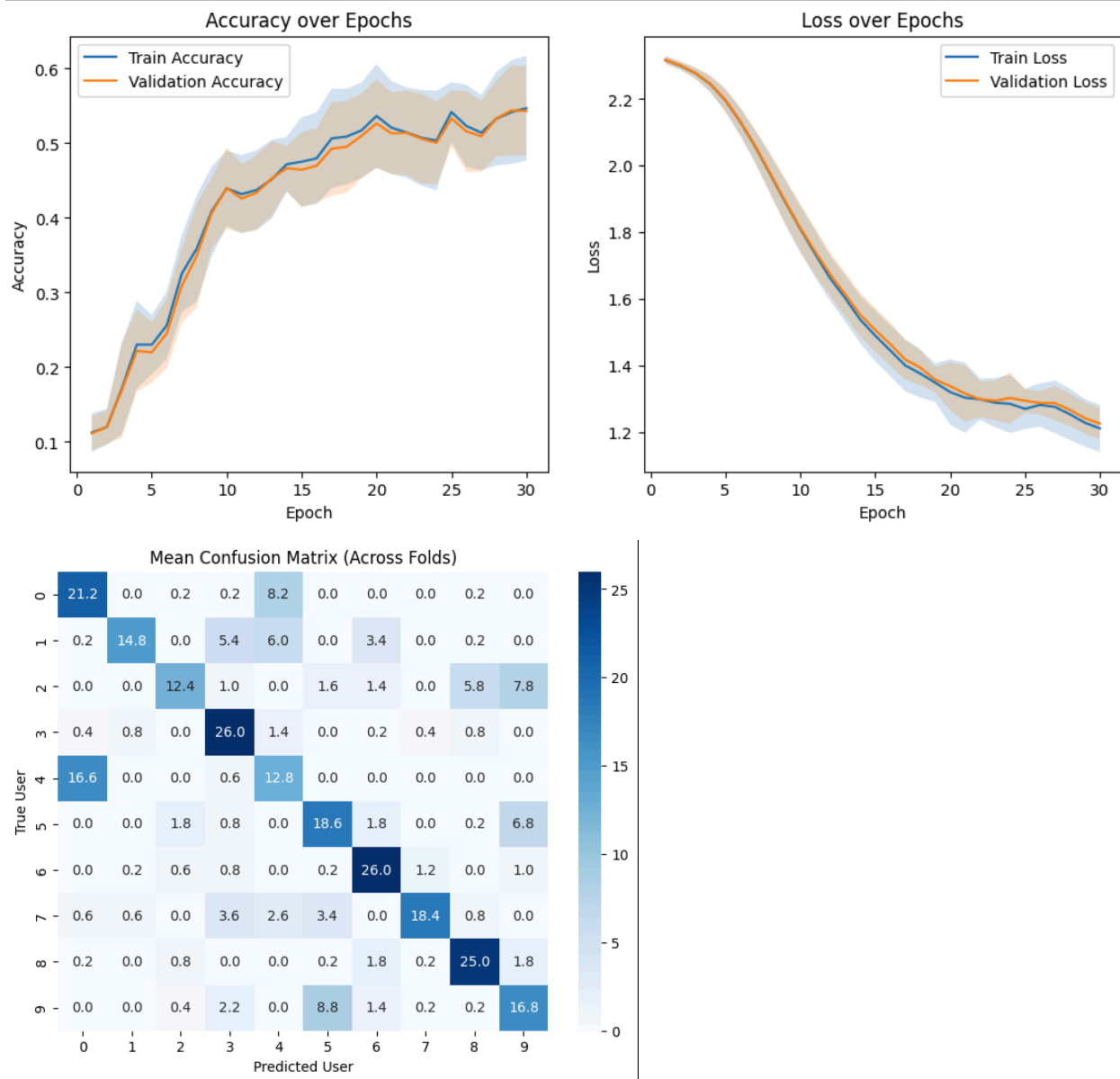
After that I cleaned the data and then I applied the moving average smoothing with a window of 5 and that helped reduce the noise without distorting the patterns making the FNN and LSTM models cleaner.

FNN Approach:

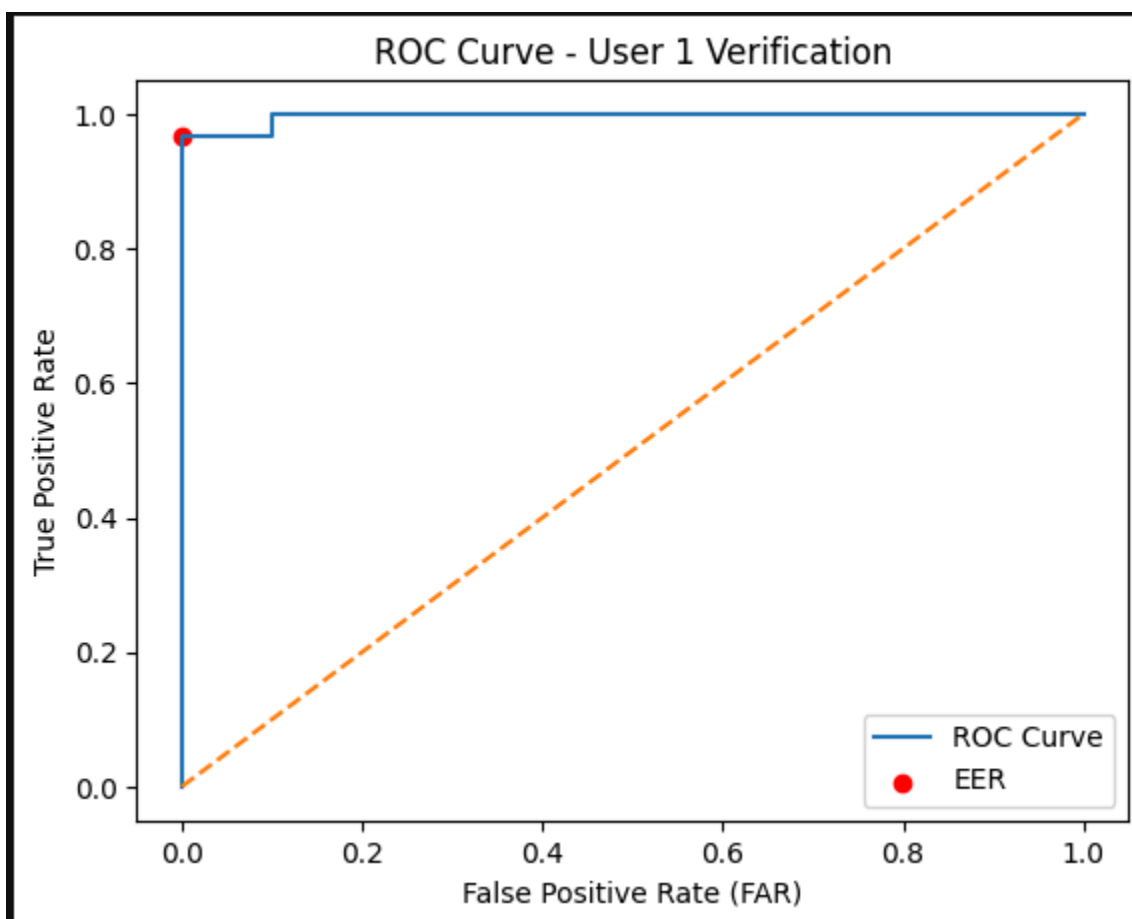
We extract features from each window segment of the six channel sensor signals. The extracted features included the mean, standard deviation, minimum, maximum, median, root mean square (RMS) and variance. After feature extraction we use feature selection to reduce the space and keep only the important features. We Used SelectKbest feature selection method that id the top features and then the final 10 features are selected to train the model.

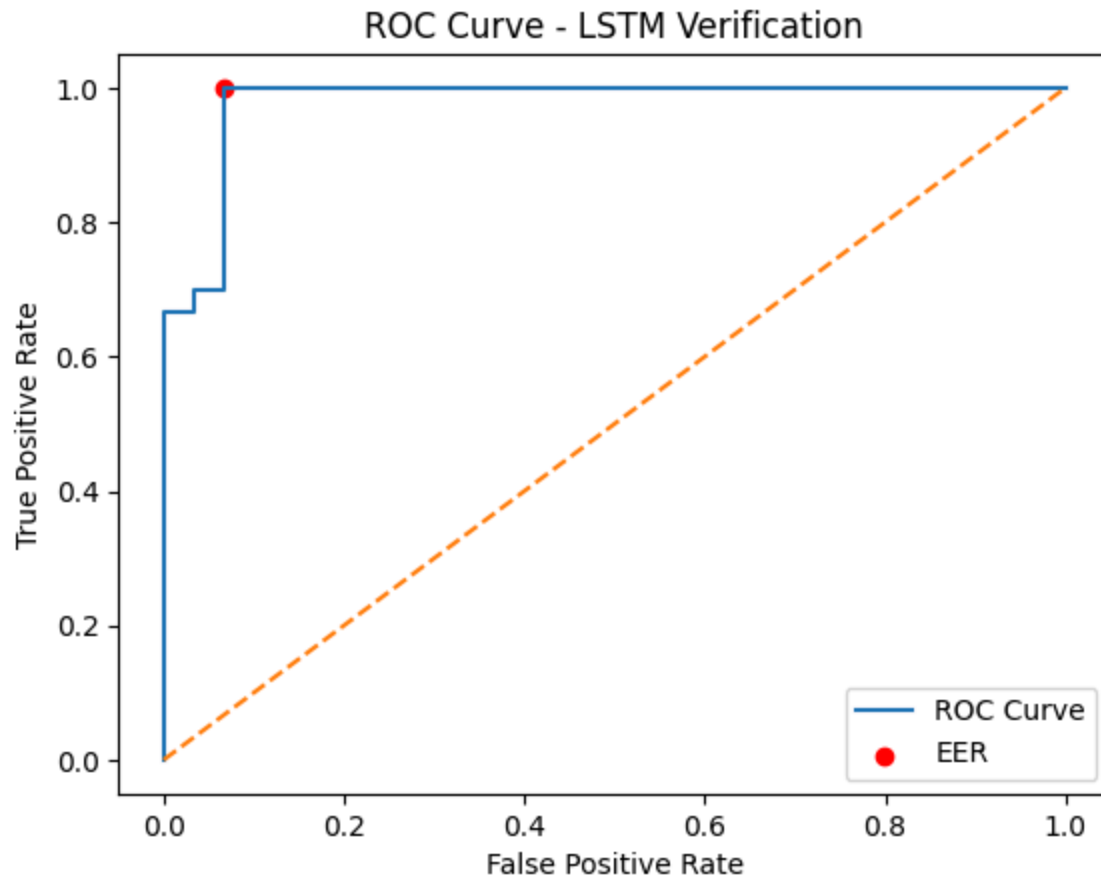
LSTM Approach:

The LSTM model processes the 6 channel window (accelerometer and gyroscope) as input and processes them using an LSTM layer with 32 hidden units to capture time-dependent motion patterns. I used Kfold cross validation, k = 5. For evaluation metrics i used recall, f1 score and also confusion matrix.



ROC curves for User 1 Verification





Discussion and Insight:

The Fnn performed better than the LSTM for the user id task. FNN got an average accuracy of 90%, while the LSTM got approximately 60-65% accuracy. This is because the FNN uses the mean, median, min and max, which also summarize the important characteristics of signals. LSTM was trained with raw signals which needed more data.

The Id task is a 10 class classification where the system must know which user made the signal. The verification was easier because it was based on binary where it's genuine and imposter for each user.

One challenge was the LSTM model because that one needed to perform more time to process the data and also class imbalances such as looking for is this user real or fake so we have to do one user vs all other users so maybe we will have user 0 with 300 samples and then fakes 1-9 have like 2700 samples so like 10% genuine and 90% fake. Model Convergence as you see during training i keep getting iteration errors.

Conclusion:

In this assignment we use gait based user id and verification using accelerometer and gyroscope data. Two models and evaluate them (FNN) using important features and (LSTM) using raw signals. The FNN got a higher accuracy of 90% and compared to LSTM about 60-65%. This means that feature extractions work well when the data is limited.